

Preserving Independent Judgment in Youth AI Use

A Study Proposal Comparing Human First AI First and Human Only Workflows

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Status *Early stage proposal. No participants have been recruited and no data have been collected.*

The question

My earlier research focused on unequal opportunity and early childhood development. It taught me that access matters, but so does what a tool or institution asks a young person to practice. As AI becomes part of schoolwork and daily life, I want to know who can use it and what habits of thought its use encourages.

AI can improve the answer we see without telling us much about what the user has learned. One student may give the right answer because she checked the AI's reasoning and came to the same conclusion. Another may give the same answer simply because she accepted what the AI said. The observed result is identical, but the reasoning behind it is not.

A useful evaluation should therefore ask two separate questions. Did AI improve the work produced while the tool was available? Did it also help the student build knowledge and judgment that remain after the tool is gone? I use independent judgment in a narrow, testable sense: performance on a related task without AI, a sound response when AI advice is wrong, and confidence that reflects actual accuracy. Recording an answer before seeing AI is an intervention, not proof that independent thought occurred.

Learning research has long distinguished performance during practice from learning that persists and transfers after practice ends (Soderstrom and Bjork, 2015). Recent experiments point to the same measurement gap without settling the design question. In a randomized field experiment with nearly 1,000 high school students, a GPT-4 interface designed to resemble ChatGPT improved performance on practice problems, but students scored 17 percent lower than the no-AI group on the unassisted exam that followed. A tutor with teacher-designed safeguards largely mitigated that loss but did not produce a positive exam effect (Bastani et al., 2025).

A preregistered study enrolled 405 students aged 14 and 15 and retained 344 after applying its exclusion criteria. Three days later, both note-taking alone and note-taking alongside an LLM improved literal retention and comprehension relative to LLM use alone. Note-taking alone also improved free recall. Students generally preferred the LLM to note-taking alone and perceived it as more helpful (Kreijkes et al., 2026). In a randomized trial with 1,053 first-year undergraduates, ChatGPT raised the score on a standard marketing rubric, while causal-reasoning training increased mechanism-based and falsifiable reasoning and the diversity of ideas but did not raise that rubric score (Asirvatham et al., 2026). Better work during AI use, learning that lasts, and the user's experience of the tool can move in different directions.

Evidence on ordering is mixed, and most of it comes from adults. In a study of 19 veterinary radiologists, participants who saw AI from the start agreed with it more often, whether the AI was correct or not. On findings with the most reliable ground truth, the one-step group performed better than the provisional-judgment group when AI was correct and worse when AI was wrong. Participants who recorded a provisional judgment first changed only 5 percent of the findings on which their initial judgment disagreed with the AI (Fogliato et al., 2022).

In three separate nutrition experiments, simulated, always-correct AI recommendations improved immediate decisions when shown before a choice and when shown after an initial choice, but neither design produced measurable incidental learning on subsequent questions in the same session. Among the AI designs tested, learning gains appeared only when the system supplied an explanation without an explicit recommendation. Because the two orders were tested in separate experiments, these results do not show that either sequence is generally better (Gajos and Mamykina, 2022). Timing matters, but so do the quality of the advice and the thinking the interface still requires from the user.

That leaves two live possibilities. AI First may improve an immediate answer by supplying useful information, but it may also pull a user toward a plausible mistake. Human First may preserve an independent starting point, but committing to an initial answer may make later revision less likely. These are hypotheses, not findings about young people. Neither tells us what young people can later do without AI. The proposed study is designed to test those questions rather than assume the answer.

A practical test

I would begin with a small feasibility study run through an institution experienced in research with adolescents. Everyone would first complete the same short task without AI. Participants would then be randomly assigned to one of three ways of working. The age range, task, setting, and sample size should follow pretesting and consultation with research partners.

Workflow	What the participant does
Human First	Sees the task, records an initial answer, confidence, and brief reason, then sees prepared AI advice and may revise.
AI First	Sees the task with the same prepared AI advice before recording an answer, confidence, and brief reason.
Human Only	Sees the task without AI, records an answer, confidence, and brief reason, and may review it independently.

The assisted groups would receive the same cases, advice, interface, and verified feedback. Some prepared advice would be correct and some would be plausible but wrong, with versions rotated across participants. Participants would know that the AI may be wrong. Using prepared advice would make the comparison clearer, although it would not reproduce a full conversation with a chatbot.

About one week later, everyone would first complete new, related tasks without AI. Only after this unaided test would the assisted groups receive additional tasks with correct and misleading advice. This order protects the main measure of what participants can do on their own.

Primary outcome Accuracy on the later tasks completed without AI.

Secondary outcomes Immediate accuracy, final decisions under correct and misleading advice, confidence relative to accuracy, explanation quality, time, perceived effort, and acceptability.

Analysis Register the main outcome and comparisons in advance, account for initial performance, use graders who do not know the assigned workflow, and report uncertainty rather than only whether a result crosses a significance threshold.

Because the participants would be minors, this should not be a solo study. It would require an institutional partner, ethics review, parental consent, youth assent, low-risk tasks, minimal data collection, corrective feedback, and a clear debrief. No identifying information should be sent to a model. Educators, young people, child development researchers, statisticians, and research ethics staff should review the tasks and procedures before recruitment.

What the evidence could and could not show

The value of the study would not be a verdict that one workflow is always safer. It would separate three questions that are often collapsed: Does AI improve today's answer? What remains when AI is gone? What happens when AI is confidently wrong? If Human First improves later unaided performance or decisions under misleading advice, a larger study could test whether products should ask users to think before showing assistance. If AI First performs as well or better, the initial answer may add effort or anchor an early mistake. If the results are similar, sequence may not be the most useful design lever. A product or school program may do well on one outcome and poorly on another, so the outcomes should be reported separately.

The study would test a whole workflow, not isolate a single psychological mechanism. Human First changes timing, effort, and commitment together. Prepared, single-turn advice is not a full chatbot experience. A result after one week would not establish long-term development, and one task, age group, or setting could not represent all youth AI use. The narrower aim is to learn whether a larger study is warranted and which outcomes it should measure.

My contribution would be evidence review, task and survey design, implementation planning, data checks, analysis, and clear documentation. The research itself should be built with partners who understand young people and the settings in which they actually use AI.

Selected references

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